

Proof of Equation (85) in the Local Limit of Nonlocal Gravity Black Holes

June 8, 2026

We wish to simplify the equation

$$\frac{d}{dr} \left(\frac{1}{U} \right) - \frac{1}{U} \left(\frac{3}{r} + \frac{S'}{1+S} \right) + 1 = 0. \quad (1)$$

where

$$U = -\frac{2r W'^2}{(W+1)(2rW' + W+1)}, \quad (2)$$

and

$$W(r) = -\frac{\mu}{2} \int_{\infty}^r \frac{du}{u^2 (1+S(u))}. \quad (3)$$

We begin by computing the derivative

$$\frac{d}{dr} \left(\frac{1}{U} \right),$$

which appears as the first term in (1). Using the chain rule, we obtain

$$\frac{d}{dr} \left(\frac{1}{U} \right) = \frac{\partial}{\partial W} \left(\frac{1}{U} \right) W' + \frac{\partial}{\partial W'} \left(\frac{1}{U} \right) W'' + \frac{\partial}{\partial r} \left(\frac{1}{U} \right). \quad (4)$$

For convenience, we define

$$\mathcal{A} = \frac{\partial}{\partial W} \left(\frac{1}{U} \right) W', \quad \mathcal{B} = \frac{\partial}{\partial W'} \left(\frac{1}{U} \right) W'', \quad \mathcal{C} = \frac{\partial}{\partial r} \left(\frac{1}{U} \right).$$

We evaluate each contribution separately. First,

$$\mathcal{A} = \frac{(1+W)(rW' + W+1)}{rW'^3} W'', \quad (5)$$

and

$$\mathcal{B} = -\frac{rW' + W + 1}{rW'}. \quad (6)$$

Finally,

$$\mathcal{C} = \frac{(1+W)^2}{2r^2W'^2}. \quad (7)$$

Next, consider the remaining terms in (1),

$$-\frac{1}{U} \left(\frac{3}{r} + \frac{S'}{1+S} \right) + 1.$$

We first express the quantity involving S in terms of W . From (3), one readily verifies that

$$\frac{S'}{1+S} = -\frac{2}{r} - \frac{W''}{W'}. \quad (8)$$

Substituting (5)–(8) into (1), we obtain an equation entirely in terms of W , W' , and W'' ,

$$\begin{aligned} \mathbf{Eq1} \implies & \frac{(1+W)(rW' + W + 1)}{rW'^3} W'' - \frac{rW' + W + 1}{rW'} \\ & + \frac{(1+W)^2}{2r^2W'^2} + \frac{(W+1)(2rW' + W + 1)}{2rW'^2} \left(\frac{1}{r} - \frac{W''}{W'} \right) + 1. \end{aligned} \quad (9)$$

We now decompose this expression into three parts:

$$\mathcal{D} = \frac{(1+W)(rW' + W + 1)}{rW'^3} W'' - \frac{(1+W)^2}{2r^2W'^2} + \frac{(W+1)(2rW' + W + 1)}{2rW'^2} \frac{W''}{W'},$$

$$\mathcal{F} = \frac{(1+W)^2}{2r^2W'^2} + \frac{(1+W)^2}{2r^2W'^2} + \frac{(W+1)(2rW' + W + 1)}{2rW'^2} \frac{1}{r},$$

and

$$\mathcal{G} = -\frac{rW' + W + 1}{rW'} + 1.$$

One can readily verify that these three quantities reproduce all terms appearing in (9).

We begin with

$$\mathcal{D} = \frac{(1+W)^2}{2rW'^3} W''. \quad (10)$$

Similarly, the second contribution simplifies to

$$\mathcal{F} = \frac{(1+W)(rW' + W + 1)}{r^2W'^2}. \quad (11)$$

At this stage, we leave $\mathcal{G}$ unchanged. Combining all contributions yields

$$\begin{aligned} \mathbf{Eq1} \implies & \frac{(1+W)^2}{2rW'^3} W'' + \frac{(1+W)(rW' + W + 1)}{r^2 W'^2} \\ & - \frac{rW' + W + 1}{rW'} + 1. \end{aligned} \quad (12)$$

The second and third terms can be combined, giving

$$\mathbf{Eq1} \implies \frac{(1+W)^2}{2rW'^3} W'' + \frac{(1+W)^2 - (rW')^2}{-(rW')^2} + 1. \quad (13)$$

The constant term $+1$ cancels, leaving

$$\mathbf{Eq1} \implies \frac{(1+W)^2}{rW'^2} \left(\frac{W''}{2W'} + \frac{1}{r} \right). \quad (14)$$

Using the identity (8), we have

$$\frac{W''}{2W'} + \frac{1}{r} = -\frac{S'}{2(1+S)}.$$

In addition, differentiating (3) gives

$$W' = -\frac{\mu}{2} \frac{1}{r^2(1+S)}.$$

Substituting these relations into (14) and simplifying, we obtain

$$\frac{d}{dr} \left(\frac{1}{U} \right) - \frac{1}{U} \left(\frac{3}{r} + \frac{S'}{1+S} \right) + 1 = -\frac{2}{\mu^2} r^3 (W+1)^2 (S+1) S'. \quad (15)$$